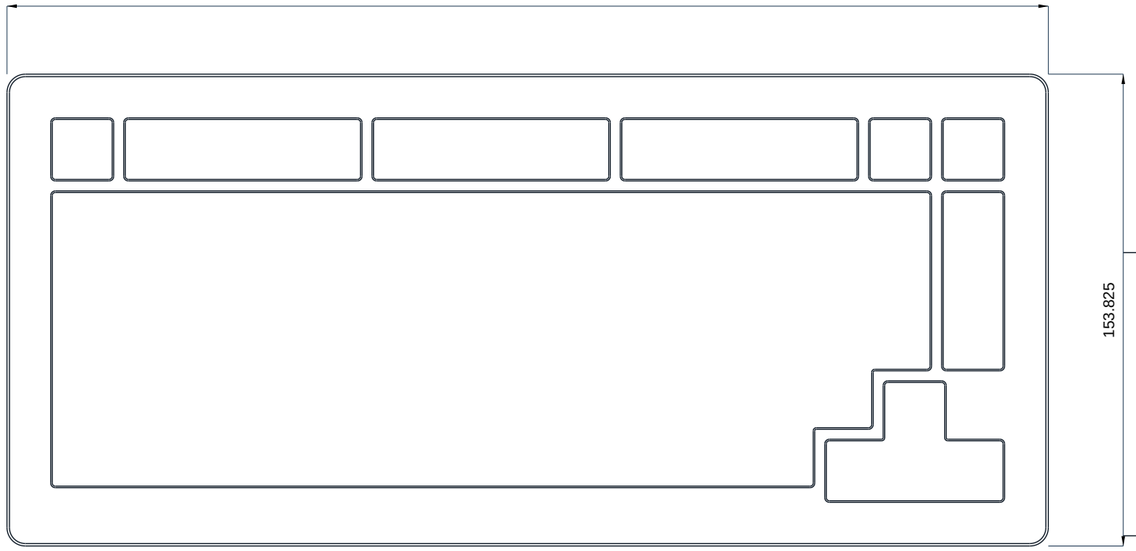
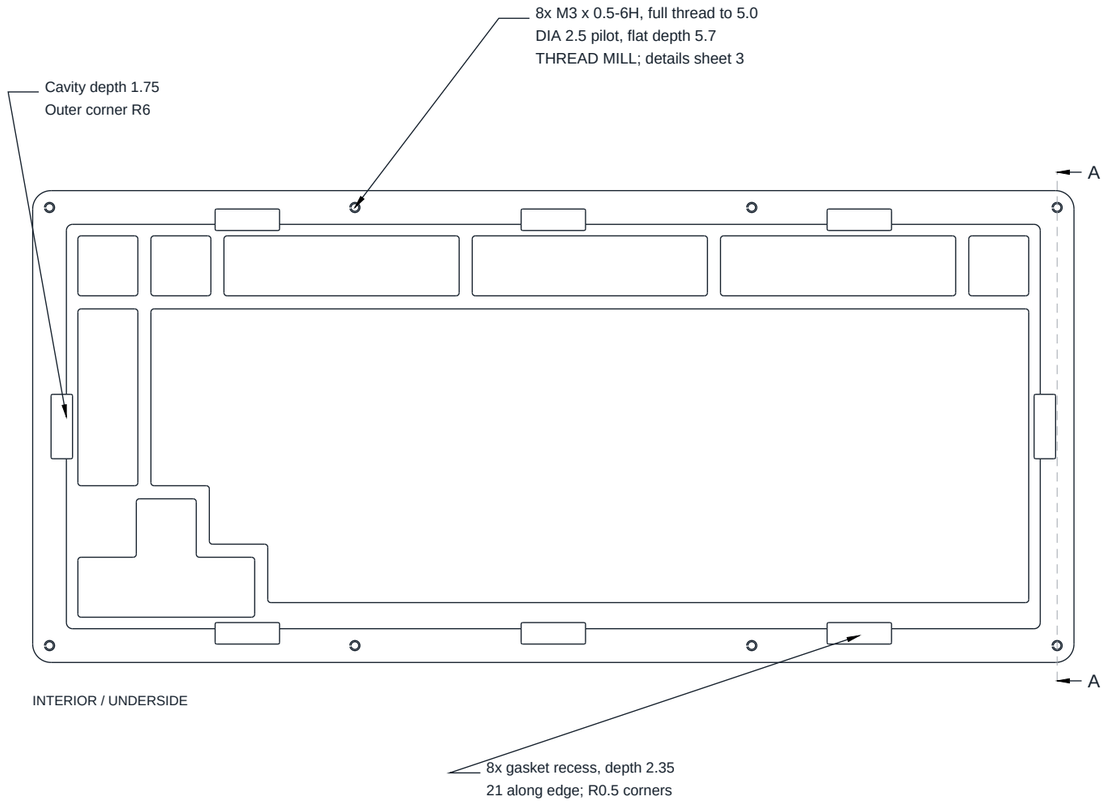


FRONT ELEVATION



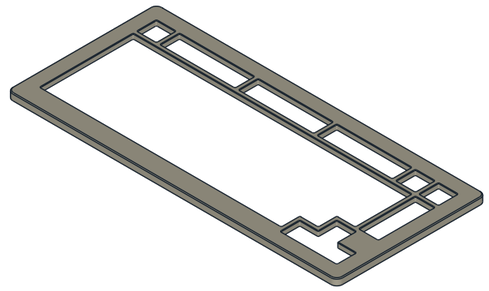
EXTERIOR / TOP



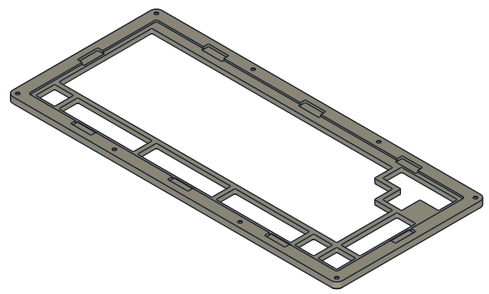
INTERIOR / UNDERSIDE



A-A (1:2)



EXTERIOR - NTS



INTERIOR - NTS

NOTES:

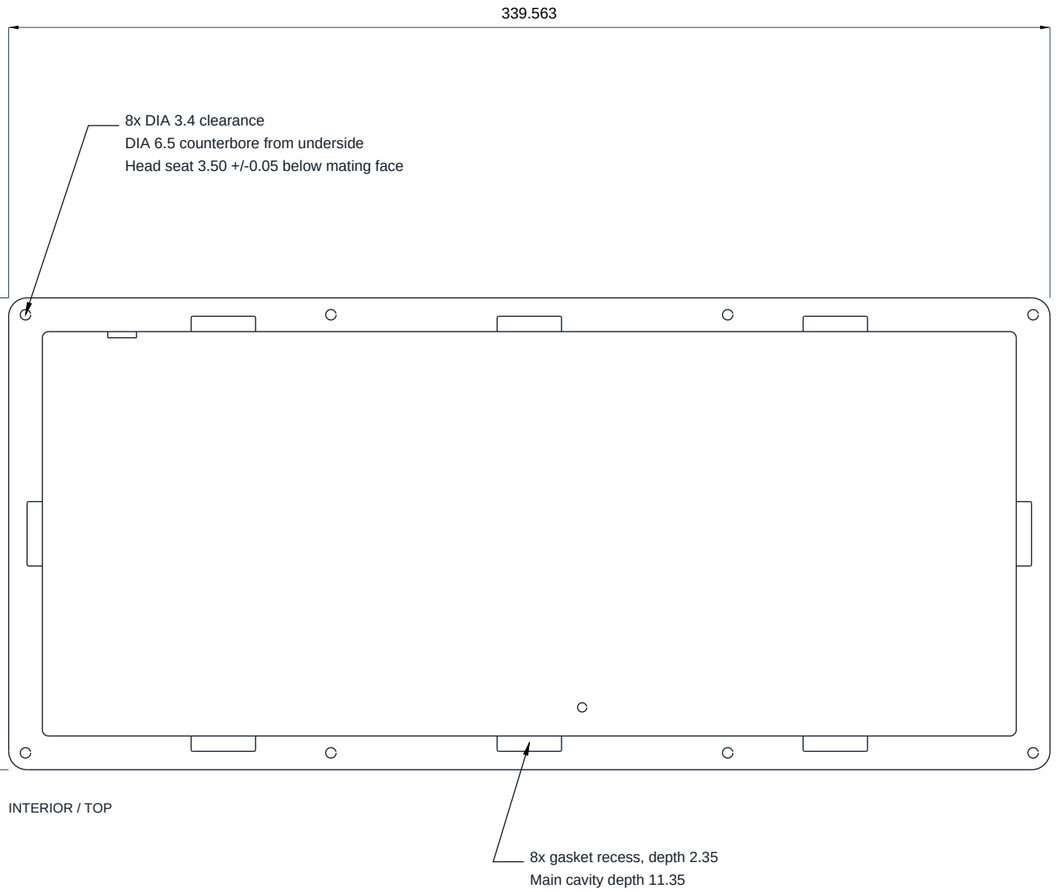
1. All dimensions in millimeters. Views 1:2; isometrics NTS.
2. General +/-0.10 mm; critical requirements on sheet 3.
3. Use with sheet 3 / CNC_REQUIREMENTS.txt. Deburr all holes.
4. Geometry per STEP. Make one pair first; total 3 pairs.
5. Key openings: R1, chamfer 0.5. Outer corners: R6.

Units	Technical reference	Prepared	Approved by
mm	Case STEP / rev 1.7	2026-11-12	--

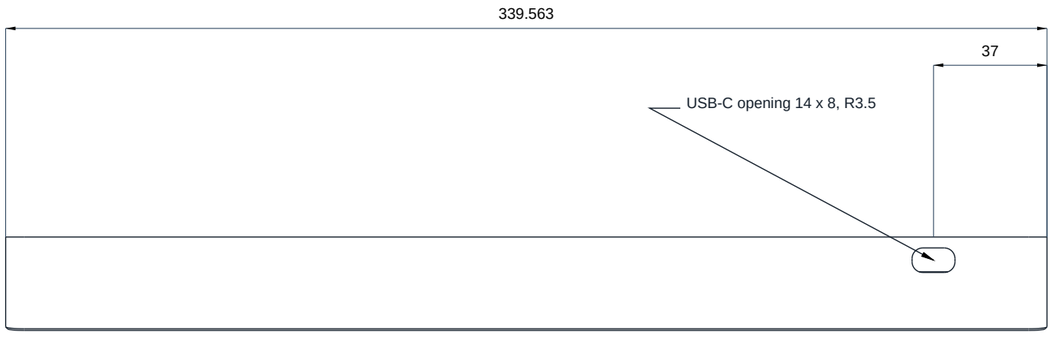
Document type	Document status
Part drawing / scale 1:2	CNC PROTOTYPE / D5

Title	Drawing No.
Vamora75 Top Case	V75-CASE-TOP

Material / finish	Rev.	Date of issue	Sheet
6061-T6 / deburred, as machined	D5	2026-11-12	1/3



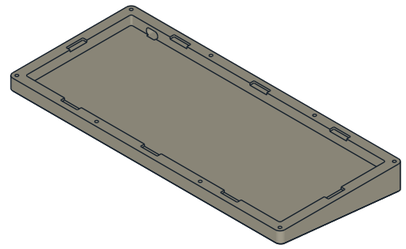
INTERIOR / TOP



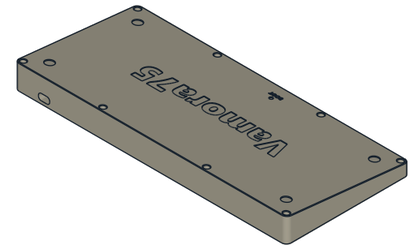
REAR ELEVATION



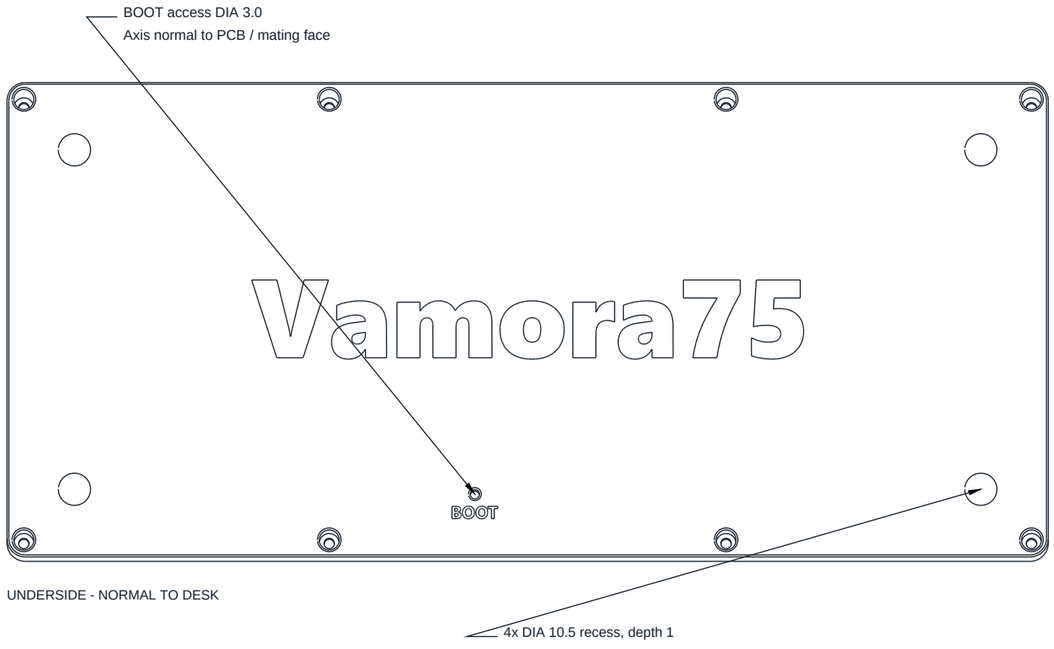
RIGHT SIDE (1:2)



INTERIOR - NTS



UNDERSIDE - NTS



UNDERSIDE - NORMAL TO DESK

NOTES:

1. All dimensions in millimeters. Views 1:2; isometrics NTS.
2. General +/-0.10 mm; critical requirements on sheet 3.
3. Use with sheet 3 / CNC_REQUIREMENTS.txt. Deburr all holes.
4. Geometry per STEP. Make one pair first; total 3 pairs.
5. Outer corners R6; bottom chamfer 0.6; engraving depth 0.6.

Units	Technical reference	Prepared	Approved by
mm	Case STEP / rev 1.7	2026-11-12	--

Document type	Document status
Part drawing / scale 1:2	CNC PROTOTYPE / D5

Title	Drawing No.
Vamora75 Bottom Case	V75-CASE-BOTTOM

Material / finish	Rev.	Date of issue	Sheet
6061-T6 / deburred, as machined	D5	2026-11-12	2/3

VAMORA75 - CNC REQUIREMENTS / D5 - SHEET 3 OF 3

Applies to both D5 STEP parts and part drawings. Dimensions in mm; no scale.

MATERIAL / FINISH / QUANTITY

6061-T6 aluminum. As machined and deburred; no anodizing or coating in this revision.

Three matched pairs total. Make and inspect one pair first, then repeat after assembly fit is confirmed.

GENERAL TOLERANCES AND DATUMS

Unspecified dimensions +/-0.10; angles +/-0.2 deg. Machined surfaces Ra 3.2 um or better.

Datum A: flat case mating face. B: left outside plane. C: front outside plane, in plate-parallel axes.

A flatness 0.10. X from B, Y from C; +Z from bottom toward top, normal to A. Desk is at 6 deg to A.

Hole coordinate table below applies to BOTH halves, viewed from above; do not mirror the bottom part.

Screw center X/Y +/-0.03 to B/C. BOOT center X/Y +/-0.10. Hole axes normal to A within 0.2 deg.

Gasket seat depth 2.35 +/-0.05 from A on each half. Keep mating faces and seats free of burrs.

TOP: BLIND THREADS - REQUIRED SECONDARY OPERATION

8x M3 x 0.5-6H. Full thread form from 0.40 to at least 5.00 into top from A; thread-mill blind holes.

STEP models DIA 2.50 pilot, FLAT depth 5.70 +/-0.05 from A; entry DIA 3.30 x 0.40 deep chamfer.

Top rim thickness 6.75 +0.10/-0.00. Max bore depth 5.75 incl. any tool tip; intact roof >=1.00.

Threads are specified here, not modeled as helices. Verify with M3-6H gauges and depth inspection.

BOTTOM: CLEARANCE / COUNTERBORES

8x DIA 3.40 +0.10/-0 through to A. DIA 6.50 +0.10/-0 counterbores open from desk underside.

Head bearing seat 3.50 +/-0.05 below A, measured along screw axis; perpendicular to that axis.

Use 8x M3 x 8 ISO 4762 screws per keyboard, head DIA 5.5 x 3, 2.5 mm hex; no washers.

Accept screw under-head length 7.80-8.20. Nominal insertion 4.50; worst-case 4.25-4.75.

BOOT / USB / EDGE DETAILS

BOOT DIA 3.00 +0.10/-0 through floor, X=187.000, Y=20.325.

BOOT axis normal to A (6 deg from desk normal). Deburr; access with a blunt 1 mm nonconductive probe.

USB rear opening 14 x 8, R3.5; center X=37.000, Z=-7.560 relative to A.

Keep all STEP reliefs and R0.5 gasket-pocket corners. Break other sharp edges 0.10-0.20 maximum.

Do not enlarge pocket radii, remove thin bridges, pocket the floor or change the logo without a revised drawing.

LAYER STACK / FIRST-ARTICLE INSPECTION

PCB 1.60; plate 1.50; plate foam 3.50; bottom foam 3.00. Use supplied foam DXFs including USB/BOOT cutouts.

16 gaskets 20 x 4.5 x 2.0 (total including adhesive), installed 1.60 per side of each of 8 tabs.

PCB underside to floor 5.50; nominal socket-to-bottom-foam gap 0.65. Foam must not preload PCB.

Inspect 8 threads, hole positions, seat depths, roof, burrs and flatness; dry-fit screws before fitting PCB.

Assemble one pair: halves meet without force; check USB insertion, BOOT press/release and all keys/stabilisers.

Digital checks are not a physical fit test. Verify gasket compression and screw engagement on the first pair.

SCREW COORDINATES (X, Y) FROM B/C; BOTH HALVES

1: X 5.500 Y 5.500

2: X 334.062 Y 5.500

3: X 5.500 Y 148.325

4: X 334.062 Y 148.325

5: X 105.081 Y 5.500

6: X 234.481 Y 5.500

7: X 105.081 Y 148.325

8: X 234.481 Y 148.325